Hoang-Chau Luong

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LLM post-training: reasoning, on-policy & knowledge distillation, unlearning

Education

Ph.D. in Computer Science, Rochester Institute of Technology — **2025 – Aug 2030 (expected)**
Golisano College of Computing and Information Sciences | Advisor: Prof. Lingwei Chen — *Rochester, NY*

B.S. in Computer Science, VNUHCM – University of Science — **2020 – 2024**
Advanced Program in Computer Science (APCS) | GPA: 3.82/4.00 — *Ho Chi Minh City, Vietnam*

Research Experience

Graduate Research Assistant — **Aug 2025 – Present**
Rochester Institute of Technology — *Rochester, NY*

- **LLM reasoning:** on-policy self-distillation for math reasoning — 1 paper under review.
- **LLM distillation:** temperature and divergence objectives for knowledge distillation — EMNLP 2026; 2 under review.
- **LLM unlearning:** making fine-tuned LLMs forget specific training data, keeping the rest — 1 paper under review.
- **Robust fine-tuning:** why parameter-efficient fine-tuning fails to remove backdoors — ACL 2026 Findings.

Publications

Conference Papers

[C1] **Hoang-Chau Luong**, Lingwei Chen. Rethinking the Role of Temperature in Large Language Model Distillation. **EMNLP 2026 (Main, Short)**.

- Temperature benefits forward KL much more than reverse KL; simple KL objectives gain up to +6.6 ROUGE-L.

[C2] **Hoang-Chau Luong**, Lingwei Chen. Why LoRA Fails to Forget: Regularized Low-Rank Adaptation Against Backdoors in Language Models. **ACL 2026 Findings**.

- Linked backdoor persistence to LoRA's limited spectral strength; RoRA lowers attack success, keeps accuracy.

[C3] **Hoang-Chau Luong**, Thuc Nguyen-Quang, Dat Ba Tran, Minh-Triet Tran. Understanding SAM's Robustness to Noisy Labels through Gradient Down-weighting. **AISTATS 2026**.

[C4] Pham Duy Khanh, **Hoang-Chau Luong**, Boris S. Mordukhovich, Dat Ba Tran. Fundamental Convergence Analysis of Sharpness-Aware Minimization. **NeurIPS 2024**.

Under Review

[P1] **Hoang-Chau Luong**, Lingwei Chen. Which Reference Did the Teacher See? Posterior-Weighted Mixture Targets for On-Policy Self-Distillation. *Under review*, 2026.

- Showed the OPSD teacher pulls correct rollouts toward the dataset's reference rather than toward correctness.
- MR-OPSD weights verified references by their posterior; improves over OPSD on all 9 model–benchmark pairs.

[P2] **Hoang-Chau Luong**, Lingwei Chen. Unlearning by Extrapolating a Shrink-and-Heal Lookahead. *Under review*, 2026.

- On TOFU (Llama up to 8B), 3.7× closer to the oracle than the next-best of 11 baselines; stable under relearning.

[P3] **Hoang-Chau Luong**, Dat Ba Tran, Lingwei Chen. Diversity-Aware Reverse Kullback–Leibler Divergence for Large Language Model Distillation. *Under review*, 2026.

- Showed reverse-KL gradients push students toward overconfidence; DRKL corrects this on Qwen3, Llama-3.2, OPT.

[P4] **Hoang-Chau Luong**, Nghia Van Vo, Kaiqi Zhao, Lingwei Chen. Consistently Informative Soft-Label Temperature for Knowledge Distillation. *Under review*, 2026.

- Sample-wise teacher/student temperatures keep soft labels informative; up to +3.6% over KD on CIFAR-100.

Technical Skills

LLM Training: PyTorch, HuggingFace Transformers, TRL, verl, vLLM, LoRA/PEFT, on/off-policy distillation
Systems: Python, C++, Linux, multi-GPU training (A100, GH200)

Honors

1st Prize, Ho Chi Minh City Mathematics Olympiad (2020) | **3rd Prize**, Ho Chi Minh City AI Challenge (2024)